Elena Favaro

Lecturer, Planetary Science

She/Her | ORCID | Google Scholar | Personal Website

EMPLOYMENT AND EDUCATION

2026 – Present Milton Keynes, UK	**Lecturer, Planetary Science, The Open University**
2024 – Present Noordwijk, NL	**Research Fellow, European Space Agency**
2020 – 2023 Milton Keynes, UK	**Postdoctoral Research Associate, The Open University**
2014 – 2020 Calgary, Canada	**Doctor of Philosophy (PhD), University of Calgary**

CV AT A GLANCE

Funding & Awards	**Eight competitive grants, awards, scholarships, and funding** totalling €1,025,000, £550,000, $178,000 (CDN), and $5,250 (USD).
Publications	**Six first-authored papers**, 11 co-authored papers, five papers in preparation.
Conference Presentations	Since 2020, I have **first-authored 13 and co-authored 37** conference and workshop presentations
Teaching and Invited Lectures	**15 invited** teaching and lecture activities totalling **21 activities**

RESEARCH EXPERTISE

My research sits at the interface of Mars surface and atmospheric science. I combine aeolian geology and geomorphology with near-surface and atmospheric wind modelling to describe how sediment transport links atmospheric dynamics to landscape evolution over diverse spatial and temporal scales. Further expertise includes:

- Martian and terrestrial fluvial and cryospheric processes, including ice–wind interactions and the distinction between modern and ancient wind regimes in mid-latitude icy terrains.
- Orbital and in situ remote sensing (HiRISE, CTX, CaSSIS, and SHARAD), GIS, terrain analysis, landing-site assessment, rover traversability, and operational geospatial products.
- Machine learning and deep learning for terrain classification, geomorphological mapping, and rover-image analysis, including automated grain-size measurement from PanCam and CLUPI imagery.
- Rosalind Franklin science and operations, including Oxia Planum geology, PanCam Science Team membership, Mars-analogue fieldwork in Argentina, Iceland, and Scotland using PanCam emulators

EXOMARS ROSALIND FRANKLIN ROVER MISSION INVOLVEMENT

2026 **Science Knowledge Program ExoMars Field Test #1**
European Space Agency
Participated in the two-week field trial at the Rover Operations Control Center (Turnin, Italy), simulating planning and execution of the ExoMars Rosalind Franklin rover stratetic and tactical plans over eight Martian sols.

2023 – Present **GRAINS - Graphical Rapid Analysis of INstrument Samples**
European Space Agency, Human and Robotic Exploration Directorate
I am the Scientific Officer and Technical Advisor for this Mission (ExoMars) Support Activity. I developed the research and technical frameworks for the project and wrote the project proposal that was endorsed by ESA's Human and Robotic Exploration Directorate and adjudicated, vetted, and awarded by the European Union Programme Board.

2022 – Present **ExoMars Rosalind Franklin Rover 'Strategic Science Plan'**
European Space Agency
'Strategic Science Plan' is the provisional title given to the largest manuscript contribution to an invited RFM *Astrobiology* issue. For this manuscript, I led a team of eight researchers in crafting the discussion for ***Question 3.4*** (section name)***: What can we learn about recent aeolian surface processes?*** I also contributed to the writing of two other sections.

2022 – Present **ExoMars Rover Simulations - Simulation Officer and Participant**
European Space Agency, The Open University
I have participated or been a Simulation Officer in seven rover simulations as a member of the PanCam team and the RSOWG.

2021 – Present **Team Member, ExoMars Rover Operations Working Group**
European Space Agency, The Open University
As a member of the RSOWG, I support scientific, programmatic, and technical aspects of the Rosalind Franklin mission.

2020 – Present **Science Team Member, ExoMars Rover PanCam Instrument**
UK Space Agency, The Open University
As a member of the PanCam science team, I support science operations, technical reviews, and testing of the PanCam emulator ahead of the 2028 launch and start of the nominal mission in 2030.

2024 **Strategic Localisation Lead, ExoMars Rosalind Franklin Simulation**
European Space Agency
Pre-simulation, I develop GIS, remote-sensing, and analogue data products and materials for simulation activities. **During the Simulation**, I coordinated activities between scientific and engineering teams, serving as the principal liaison to ensure scientific objectives were accurately translated into operational plans and field execution.

SELECTED COMPETITIVE GRANTS, AWARDS, AND FUNDING

2025 **Faculty Research Funding Grant (Ongoing)**
European Space Agency, Science Directorate
Title of research project: *Deciphering Mars' Wind and Icy Landscapes with Accessible Machine Learning Models* **(€25,000)**

2024 **GRAINS - Graphical Rapid Analysis of INstrument Samples (Ongoing)**
European Space Agency, Human and Robotic Exploration Directorate
Funding secured at ESA for mission support activities (more information is provided in the ExoMars Rosalind Franklin Rover Mission Involvement section above) **(€250,000)**.

2023 **ESA Research Fellowship in Space Science (Current Position)**
European Space Agency
Proposal: Developing Accessible Machine Learning Techniques to the study of Martian Aeolian Processes **(€750,000)**.

2023 **Leverhulme Early Career Fellowship at University of Sheffield**
Leverhulme Trust
Awarded **£550,000**. Alternative Fellowship taken up.

2021 **Fiona Kirkby Award**
British Society for Geomorphology
In "recognition of the outstanding support… to the journal, but also its authors through reviews that are timely, detailed and constructive"

2016 **Izaak Walton Killam Pre-Doctoral Scholarship (2016-2018)**
University of Calgary
Awarded to top Graduate Student Scholars at the University of Calgary **($108,000 Canadian)**

2016 **Graduate Student Award for Outstanding Contributions to Community Life**
Graduate Student Association, University of Calgary

2016 **Farouk El-Baz Student Research Grant**
Geological Society of America
Awarded to a student engaged in desert-related research **($2,500 USD).**

2015 **Geological Society of America Research Grant (1st Tier)**
Geological Society of America
Awarded to graduate students undertaking fieldwork as part of their graduate research **($2,750 USD)**

2015 **Natural Sciences and Engineering Council of Canada (NSERC) Postgraduate Scholarship – Doctoral (2015-2018)**
Natural Sciences and Engineering Council of Canada
Awarded to top doctoral students in Canada **($60,000 Canadian).**

PUBLICATIONS

2026 Grindrod, P. M., **Favaro, E. A.**, Balme, M. R., & Patel, M. R. (2026). Active Dust Devils in the Rosalind Franklin Rover Landing Site in Oxia Planum, Mars. *Journal of Geophysical Research: Planets*, *131*(10), e2026JE010056.

2026 Barrett, A. M., Balme, M. R., **Favaro, E. A.**, Woods, M. J., Malinowski, M., Elsarky, L., ... & Patel, M. R. (2026). Detecting and measuring Transverse Aeolian Ridges on Mars using deep learning. *Icarus*, 116947.

2024 Fawdon, P., Orgel, C., Adeli, S., Balme, M., Calef, F. J., Davis, J. M., ... **Favaro, E. A.**, ... & Zorzano, M. P. (2024). *The high-resolution map of Oxia Planum, Mars: the landing site of the ExoMars Rosalind Franklin rover mission.* Journal of Maps, 20 (1), 2302361.

2024 **Favaro, E.A.**, Balme, M.R., Davis, J.M., Grindrod, P.M., Fawdon, P., McNeil, J.D., & Lewis, S.R. (2024). *Periodic Bedrock Ridges at Oxia Planum and Chryse Planitia, Mars: Evidence for widespread aeolian erosion of an ancient surface by regional palaeowinds.* Earth and Planetary Science Letters, 626, 118522.

2023 Barrett, A. M., Fawdon, P., **Favaro, E. A.**, Balme, M. R., Wright, J., Woods, M. J., ... & Joudrier, L. (2023). Mawrth Vallis, Mars, classified using the NOAH-H deep-learning terrain classification system. *Journal of Maps*, 19(1), 2285480.

2023 Davis, J.M., Balme, M.R., Fawdon, P., Grindrod, P.M., **Favaro, E.A.**, & Banham, S.G. (2022). Ancient Alluvial Plains at Oxia Planum, Mars. *Earth and Planetary Science Letters,* 601, 117904.

2023 Barrett, A. M., Wright, J., **Favaro, E.**, Fawdon, P., Balme, M. R., Woods, M. J., ... & Joudrier, L. (2023). Oxia Planum, Mars, classified using the NOAH-H deep-learning terrain classification system. *Journal of Maps*, *19*(1), 2112777.

2022 Wright, J., Barrett, A. M., Fawdon, P., **Favaro, E. A.**, Balme, M. R., Woods, M. J., & Karachalios, S. (2022). Jezero crater, Mars: application of the deep learning NOAH-H terrain classification system. *Journal of Maps*, *18*(2), 484-496.

2021 **Favaro, E. A.**, Hugenholtz, C. H., & Barchyn, T. E. (2021). Antecedent controls on the spatial organization of yardangs on the Puna Plateau, north-western Argentina. *Earth Surface Processes and Landforms,* 46(15), 3063-3077.

2021 **Favaro, E. A.**, Balme, M. R., Davis, J. M., Grindrod, P. M., Fawdon, P., Barrett, A. M., & Lewis, S. R. (2021). The Aeolian Environment of the Landing Site for the ExoMars Rosalind Franklin Rover in Oxia Planum, Mars. *Journal of Geophysical Research: Planets,* 126(4), 2020JE006723.

2020 **Favaro, E. A.**, Hugenholtz, C. H., Barchyn, T. E., & Gough, T. R. (2020). Wind regime, sediment transport, and landscape dynamics at a Mars analogue site in the Andes Mountains of Northwestern Argentina. Icarus, 346, 113765.

2017 **Favaro, E. A.**, Hugenholtz, C. H., & Barchyn, T. E. (2017). Evolution and diagnostic utility of aeolian rat-tails: A new type of abrasion feature on Earth and Mars. Aeolian Research, 28, 91-98.

2017 Bonnaventure, P. P., Lamoureux, S. F., & **Favaro, E. A.** (2017). Over-Winter Channel Bed Temperature Regimes Generated by Contrasting Snow Accumulation in a High Arctic River. Permafrost and Periglacial Processes, 28(1), 339-346.

2015 **Favaro, E. A.**, & Lamoureux, S. F. (2015). Downstream patterns of suspended sediment transport in a High Arctic river influenced by permafrost disturbance and recent climate change. Geomorphology, 246, 359-369.

2015 Hugenholtz, C. H., Barchyn, T. E., & **Favaro, E. A.** (2015). Formation of periodic bedrock ridges on Earth. *Aeolian Research*, 18, 135-144.

2014 Lamoureux, S. F., Lafrenière, M. J., & **Favaro, E. A.** (2014). Erosion dynamics following localized permafrost slope disturbances. *Geophysical Research Letters,* 41(15), 5499-5505.

2008 Percy, J. R., **Favaro, E. A.**, Glasheen, J., Ho, B., & Sato, H. (2008). Period changes in pulsating red supergiant stars: a science and education project. *Journal of the American Association of Variable Star Observers (JAAVSO)*, 36, 145.

In Prep **Favaro, E, A.**, Balme, M.R., McNeil, J.D., Davis, J.M., Fawdon, Banham, S.G., P. Grindrod, P.M., Lewis, S.R., Rajendran, K., Holmes, J.A. and Patel, M.R. *Periodic Bedrock Ridges across Equatorial Mars: Insights from a Synoptic Survey.* In Review at JGR Planets.

In Prep Patel, M.R., **Favaro, E.A.**, Rajendran, K., Holmes, J.A., Fawdon, P. *Matching and interpretation of Martian wind streaks using mesoscale modelling.* In Prep for Earth and Planetary Science Letters.

In Prep Jackson, D.W.T, Byers, M., **Favaro, E.A.**, Balme, M.R., Lewis, S.R. Hidden Winds on Mars: Evidence for widespread mobilization of loose sediments. In Prep for Nature Geoscience.

In Prep Banham, S.G., **Favaro, E.A.**, Carling, P., Balme, M.R. Trenching of mafic gravel megaripples, Askja, Iceland: Insights to the internal architecture of extra-terrestrial megaripples. In Prep for Aeolian Research.

In Prep Banham, S.G., **Favaro, E.A.** Deciphering sedimentary processes, architecture, and stratigraphy of a Proterozoic deltaic succession as an analogue for Martian deltas: the Cailleach Head Formation, NW Scotland. In Prep for Geology.

SELECTED CONFERENCE PRESENTATIONS

2026 **Favaro, E. A.** (2026). Every Grain Tells a Story: Deciphering Planetary Processes Through Sediment Transport on Earth, Mars, and Beyond. In *Europlanet Science Congress (EPSC2026-655).*

2026 **Favaro, E. A.**, Fernandez, L., Fayolle, S., Barrett, A., Balme, M. R., Fawdon, P., Wright, J., and Joudrier, L. (2026). Developing Flexible Algorithms to Optimize Drive Paths for the ExoMars Rosalind Franklin Rover. In *European Geophysical Union Conference (EGU26-12037).*

2025 **Favaro, E. A.**, Patel, M. R., Rajendran, K., and Holmes, J. A. (2025). A Mesoscale Modelling Approach to Understanding Mars' Aeolian Landscapes. In *Europlanet-Division for Planetary Science Joint Meeting, EPSC-DPS Joint Meeting (EPSC-DPS2025).*

2024 **Favaro, E. A.**, Balme, M. R., McNeil, J. D., Fawdon, P., Davis, J., Grindrod, P. M., ... & Lewis, S. R. (2024). Insights from an Equatorial Survey of Periodic Bedrock Ridges on Mars. In *Tenth International Conference on Mars (Vol. 3007, p. 3263).*

2024 **Favaro, E.A.**, Balme, M.R., McNeil, J.D., Fawdon, P., Davis, J., Grindrod, P.M., Banham, S., Lewis, S.R. (2024). Periodic Bedrock Ridges across the equatorial region of Mars: Insights from a global survey, in *British Planetary Science Conference, Leicester, UK*

2023 **Favaro, E.A.,** Balme, M.R., Fawdon, P., Davis, J., Barrett, A.M., McNeil, J.D., Grindrod, P.M., Lewis, S.R. (2023). Oxia Planum, Mars: A history of Aeolian activity as told by bedforms and landforms. In *Fluvial Aeolian Interactions on Planetary Surfaces (FAIRPLAY), ESA/ESTEC, Noordwijk, NL.*

2023 **Favaro, E. A.**, Balme, M. R., Ladegaard, A., Gunn, M., Jackson, D. W. T., Rogers, D., & Banham, S. G. (2023). Preliminary Results from a Comparison Between the ExoMars PanCam Emulator and Terrestrial Laser Scans of Icelandic Megaripples. In *54th Lunar and Planetary Science Conference (Vol. 2806, p. 2204).*

2022 **Favaro, E.**, Balme, M., Davis, J., Grindrod, P. M., Fawdon, P., McNeil, J. D., ... & Lewis, S. (2022). A Characterization of the Aeolian Environment at Oxia Planum, Mars. In *Geological Society of America Abstracts (Vol. 54, p. 380005).*

2022 **Favaro, E. A.**, Balme, M. R., Davis, J., Grindrod, P. M., Fawdon, P., McNeil, J. D., ... & Lewis, S. R. (2022). Using Granular Bedforms and Bedrock Landforms to Characterize the Aeolian Environment at Oxia Planum, Mars. In *7th International Planetary Dunes Workshop (Vol. 2682, p. 3018)*.

2022 **Favaro, E.A.**, Balme, M.R., Davis, J., Grindrod, P.M., Fawdon, P., McNeil, J.D., Barrett, A.M., Lewis, S.R. (2022). The Aeolian Environment of Oxia Planum, In *British and Planetary Science Conference, Milton Keynes, UK.*

2022 **Favaro, E. A.**, Balme, M. R., Davis, J., Grindrod, P. M., Fawdon, P., McNeil, J. D., ... & Lewis, S. R. (2022). Using Granular Bedforms and Bedrock Landforms to Characterize the Aeolian Environment at Oxia Planum, Mars. *7th International Planetary Dunes Workshop,* LPI Contributions, 2682, 3018.

2022 **Favaro, E. A.**, Balme, M. R., Davis, J., Fawdon, P., Grindrod, P. M., McNeil, J. D., & Lewis, S. R. (2022). Periodic Bedrock Ridges on Clay-Bearing Terrain at Oxia Planum, Mars. In *52nd Lunar and Planetary Science Conference (Vol. 2678, p. 2259).*

2020 **Favaro, E. A.**, Balme, M. R., Davis, J., Grindrod, P. M., Fawdon, P., Barrett, A. M., & Lewis, S. R. (2020). Aeolian Signals from landforms and bedforms at the Landing Site for the ExoMars Rosalind Franklin Rover in Oxia Planum, Mars. *In Europlanet Science Congress (No. EPSC2020-572)*.

2020 – Present I coauthored **37 abstracts at seven national and international conferences and workshops**

2012-2019 I presented at **10 national and international conferences**

SELECTED PROFESSIONAL LEADERSHIP AND SERVICE

Present **MEDaL-1 Science Definition Team & Regolith Subgroup Chair**
European Space Agency, Human and Robotic Exploration Directorate
Selected to the ESA Science Definition Team responsible for defining the scientific rationale, measurement strategy, payload requirements, and science traceability of the MEDaL-1 (Mars Entry, Descent, and Landing) Mars lander mission. Contribute expertise in Mars atmosphere-surface coupling, dust and regolith processes, landing-site science, and science-engineering translation to support mission definition and study-phase trade-offs.

2024 – 2026 **European Space Agency Concurrent Design Facility (CDF) Activities**
European Space Agency, Human and Robotic Exploration Directorate
MEDaL (Mars Entry Descent and Landing; 2024-2026): I was responsible for **remote sensing support and GIS analysis** and selection of landing sites, adhering to operational/engineering constraints. I was responsible for **creating cartographic and digital elevation products**, as well as **all spatial analyses**. I was tasked with liaising with engineering teams to translate science requirements and goals into systems engineering protocols. I also contributed to writing the final CDF report.

To support upcoming Phase A activities, I am reassessing landing site selection and providing geospatial analyses and products to engineering and industrial teams.

Lightship-2 (2025): I was an informal participant in the Lightship-2 CDF, **providing remote sensing and GIS analysis support** for landing site selection for the primary customer team, as well as **creating cartographic products** used during the activity.

2022 – Present **Girl Guides of Canada National Awards Reviewer**
Girl Guides of Canada
Adjudication of national award applications from across Canada

2020–Present **Mentor, Europlanet Mentorship Programme**
Europlanet Society

2016–Present **Reviewer**
Earth Surface Processes and Landforms; Earth and Planetary Science Letters; Aeolian Research; Icarus; Geomorphology; Journal of Geophysical Research: Planets

2024 – 2026 **Research Fellow Application Adjudication**
ESA Science Directorate
I serve on the adjudication committee tasked with ranking all eligible Research Fellow applications.

2024 – 2025 **Chair of the Local Organising Committee for the 17th Space Science Workshop, Science Directorate, ESA**
European Space Agency

2025 **ESA/NASA Early Careers Panel**
Europlanet Early Careers at Europlanet Science Congress
I was the only early career researcher and only woman panellist for an early career event at EPSC, aimed at informing ECRs about working at ESA and NASA.

2021 – 2024 **Earth Surface Processes and Landforms Advisory Board Member**
Earth Surface Processes and Landforms

2023 **Organiser, ArcGIS Pro and ArcGIS Online Workshop**
The Open University
Liaised with colleagues at Esri UK to facilitate a full-day in-person and online workshop for PhD students, research staff, and lecturers from 6 institutions across the UK. Developed educational material with Benedek Marozsák (Esri UK).

2020 – 2022 **Conference Organiser and Co-chair, Organising and Technical Committees**
British Planetary Science Conference
Organiser of BPSC 2022. Solely responsible for website design and implementation, managing social media accounts, managing conference inbox, and all conference volunteers (~50). Co-developed abstract submission forms, managed submissions, evaluations, and communication with the scientific community.

2018 – 2021 **Session Advocate & Organiser - 'Geomorphology of Mars' and 'Analogue Surface Studies' sessions**
Geological Society of America Annual Meeting

2017 – 2019 **Session Advocate & Organiser - 'Martian and Terrestrial Aeolian Processes' session**
American Geophysical Union Fall Meeting

2014 – 2019 **Lead Organiser/Conference Chair, Multiple Conferences**
University of Calgary
From 2014 to 2019, I was the lead organiser for one national conference (2015; **11th ACUNS Student Conference**), one university-wide graduate student conference (2019; **3rd Annual Peer Beyond Graduate Research Symposium**), and two **Department of Geography conferences** (2015, 2016). I led organisational and technical teams, developed programming, built informational websites and online platforms for abstract submissions, liaised with external stakeholders, secured funding and sponsorships, invited speakers, and managed on-site activities.

Dates Redacted **Grant Reviewer - PROdex for Science in Space Exploration**
Portuguese Space Agency
Expert reviewer for the Portuguese Space Agency's call for proposals about Martian geology, geomorphology, and atmospheric science.

Dates Redacted **NASA Review Panel Executive Secretary**
NASA

SELECTED PROFESSIONAL TRAINING

Present — **Inclusive Leadership Training**
The Open University (OpenLearn)
Enrolled in the first of a three-course module - **Inclusive Leadership: Effecting Change**

2021 — **Introduction to EDI (Equity, Diversity, Inclusion)**
The Open University and Advance HE
A 24-hour 3-course module delivered online, requiring weekly submitted assignments, online discussion, and coursework.
- Introduction to EDI (Equity, Diversity, Inclusion) in Teaching and Learning
- Inclusive Curriculum Development and Delivery
- Being an EDI Change Agent

2018 – 2019 — **Graduate Student Certificate in Teaching and Learning**
University of Calgary
A two-year, 60+ hour voluntary program, consisting of 5 modules:
- Learning Spaces and Digital Pedagogy Program: Graduate Students
- Theories and Issues in Postsecondary Teaching and Learning: Graduate Students
- SoTL Foundations Program: Graduate Students
- Emerging Teachers Development: Graduate Students
- Developing Your Teaching Dossier: Graduate Students

2016 – 2019 — **Student Wellness Services Training**
University of Calgary
Specialised mental health and intervention trainings (various durations):
- ASIST: Applied Suicide Intervention Skills Training
- Suicide Prevention Training
- Responding to Students in Distress
- Bystander Awareness Training

SELECTED SCIENCE COMMUNICATION AND OUTREACH ACTIVITIES

2024 – Present — **Editorial Support**
Various ESA Publications
I provide mapping, image selection, and editorial input for Mars-related posts on ESA's **To Mars and Back** blog (Craters on Mars – To Mars and back ; Mermaids, old ponds and smileys – To Mars and back ; Where to go on Mars, Rosalind? – To Mars and back). and ESA Multimedia (ESA - Crater in Valles Marineris ; ESA - Impact crater on Mars ; ESA - Metallic Mars).

2021 – Present — **Expert Commentator**
Various Outlets
Eos Magazine, Commentary on Yardang formation and the development of Mud Lions. 7 November 2022; **Physics World**, "Simulations show that asteroid impacts would destroy evidence for relic shorelines on Mars". 30 June 2022; **Nature News**, "China's Mars rover returns first images — scientists say the view is promising". 20 May 2021.

2020 – Present

Invited Public Outreach Lectures (Selected)
Canada, Europe, UK
Galway Geological Association (UK); International Women and Girls in Science Day (Canada & UK), STEM Day (Canada); ArcticNet (Canada); Basel University (Switzerland); Scientists: What Do They Look Like? (Canada); Scientists: What Do They Do? (Canada).

2025

Invited Panelist
Space Rocks Education: Your Ticket to Space
Invited panellist for ESA's Open Day education outreach session.

2025

ESA and Research Fellow Ambassador
Europlanet Science Congress
I co-presented presentations on how to apply for the ESA Research Fellowships in the Science Directorate, and what being a Research Fellow entailed.

2024

Planetary Scientist Liaison
ESA Open Day
During ESA's 2024 Open Day, I interacted with thousands of visitors, discussing Mars missions and associated science, as well as astronomical, astrophysical, and heliophysical science.

2016 – 2022

STEM Day Coordinator
Calgary Area Girl Guides, Girl Guides of Canada
I led an interdisciplinary team (2016-2019) developing engaging scientific outreach materials for girls aged 5-17. This included interactive activities like science scavenger hunts, chemistry demonstrations, building 3D models with handheld scanners, as well as explaining complex scientific concepts through Lego building, songs, and games. The materials we developed are still being used today. These activities were adapted for online delivery between 2020 and 2022.

TEACHING AND INVITED LECTURES

2024

Guest Seminar
Basel University (Switzerland)
The Aeolian Environment at Oxia Planum: Supporting Rover Operations through Terrestrial Analogues, Atmospheric Modelling, and High-Resolution Orbital Observations.

2022

Guest Speaker (Webcast)
The Open University
Guest speaker in S283/284: Planetary Science and the Search for Life/Astronomy.
Planets and Moons Chat (https://tinyurl.com/webcastOU)

2022

Guest Lecture
GRADNet Astrobiology and Planetary Science Workshop (Remote; UK).
The Physics of Sediment Transport on Mars (And Why We Should Care)

2021 **Guest Lecture (Hybrid)**
University of Lethbridge
Geography 3300A: Microclimatology. Winds and Microclimates of Earth and Mars: Microclimatology at the 2023 Landing Site of the European Space Agency's ExoMars Mission

2021 **Guest Lecture and Mars Yard Facilitator**
The Open University
Guest speaker in S808: Space Science. The Physics of Sediment Transport on Earth and Mars

2021 **Guest Seminar**
ESA Rover Science Operations Working Group (Remote)
The Aeolian Environment of Oxia Planum: What Rosalind Franklin Might Find (and why it's important we look)

2021 **Guest Seminar**
South East Physics Network (UK)
Modern Mars: A (Brief) Exploration of a Dynamic World

2020 **Guest Lecture (Remote)**
University of Calgary (Canada)
Developed a lecture for Geography 280: Thinking Spatially in a Digital World. *Applying GIS Techniques to the Study of Mars*

2019 **Guest Lecture Series**
University of Calgary (Canada)
Developed two lectures for Geography 201: Physical Geography. *A Tale of Two Planets: Geomorphology of Earth and Mars.*

2018 **Guest Lecture**
University of Calgary (Canada)
Developed a lecture for Geography 205: Gateway to Geography. *Atmospheric Hazards: Anatomy of Cyclones*

2018 **Guest Lecture**
University of Calgary (Canada)
Developed a lecture for Geography 437: Cartography and Geographic Visualization. *A Cartographic Trip Through the Solar System*

2017 **Guest Lecture**
University of Calgary (Canada)
Developed a lecture for Geography 415: Hydrology. *So You Want to Be a Hydrologist? Experiences from the Arctic*

2017 **Guest Lecture Series**
University of Calgary (Canada)
Developed three lectures for Geography 207: Wind Science: *Wind Interaction with Mountain Topography*; *Aeolian Processes: An Introduction*; *Aeolian Processes in our Solar System*

2014 **Guest Lecture and Laboratory Series**
Queen's University (Canada)
Developed two lectures and one laboratory assignment for Geography 312: Watershed Hydrology.

2013 **Guest Lecture Series**
Queen's University (Canada)
Developed three lectures for Geography 208: Surface Processes, Landform, and Soils.

FIELDWORK EXPERIENCE

I have **over 4 months** (combined) of fieldwork experience in remote locations in a variety of different environments conducting geology, geomorphology, hydrology, and aeolian/atmospheric research. Since 2011, I have been responsible for fieldwork planning.

Over 14 days in 2022, I conducted novel atmospheric, geologic, and geomorphologic research using AUPE (Aberystwyth University PanCam Emulator) at **Lake Askja, Iceland.** This work directly supports rover and instrument operations for the **ExoMars Rosalind Franklin Rover** mission, and **PanCam** instrument deployment.

Other fieldwork campaigns include:

2021 (14 days): Scoraig, Scotland: I assisted in geologically mapping delta successions on Scoraig, the data of which are being used to compare to *in situ* observations from MSL Curiosity of possible delta successions in Gale Crater.

2015 and 2016 (12 days): Puna Plateau, Catamarca Province, Argentina: As a **PhD student researcher**, I proposed, planned, and conducted *in situ* aeolian geomorphology and geology research in a Martian analogue field site. My fieldwork was **practically supported** by **remote sensing datasets** of the field site and by **high-resolution Mars remote-sensing products**, and **financially supported** by two independent research grants awarded by the Geological Society of America.

2013 (35 days): Cape Bounty Arctic Watershed Observatory, Melville Island, Nunavut: As **the sole camp hydrologist**, I installed and maintained all hydrological monitoring equipment, conducted lake and river water and chemistry sampling and analysis, **supervised** a Master's student and undergraduate student researcher, and was one of two senior researchers responsible for camp safety.

2011 (66 days): Cape Bounty Arctic Watershed Observatory, Melville Island, Nunavut: As a **Master's student researcher**, I proposed, planned, and conducted fluvial geomorphology and hydrology research on a remote island in the Canadian Archipelago. I installed and maintained monitoring equipment, and conducted novel research that utilised remote sensing, in situ sample collection and analysis (e.g. mercury and heavy metal analysis), and high-resolution survey mapping of fluvial channels and permafrost exposure and extents.

2009 (10 days): Bowling Green, Kentucky: **As an undergraduate student researcher**, I conducted and supported hydrological, geomorphological, and sedimentological research in Mammoth Cave with Dr Chris Smart at the University of Western Ontario.